Open-Source RAG System — Evaluation Report

RBI Document QA · qwen2.5:7b · BAAI/bge-base-en-v1.5 · BAAI/bge-reranker-base · Local Ollama · MLflow 3.17.2

1. System Overview

RAG pipeline over RBI Master Directions. Documents chunked at 512 tokens with 64-token overlap; FAISS + BGE embeddings; optional BGE cross-encoder reranking. Local generation and LLM judge use qwen2.5:7b via Ollama. Benchmark: 40 questions (26 retrieval-labeled, 8 unanswerable, 6 ambiguous).

2. Retrieval: Baseline vs. Reranking (n=40)

Configuration	Recall@5	Recall@10	MRR@10	Latency context
Baseline (FAISS only)	0.769	0.885	0.648	Retrieval baseline
+ BGE reranker (4× candidates)	0.962	0.962	0.862	Retrieval-only run: ~18.6 s mean

Reranking improves Recall@5 by 19.23 percentage points and MRR@10 by 21.42 percentage points. Retrieval-only latency should not be compared directly with full-RAG latency.

3. Full-RAG Answer Quality & Abstention

Metric	Measured result	Interpretation / caveat
Baseline LLM-judge correctness / faithfulness	0.765 / 0.894	34 judged; 6 ambiguous questions skipped. Same model used for generation and judging, so self-evaluation bias applies.
Exact-substring correctness (baseline)	0.115	Naive lexical heuristic; not semantic accuracy.
Targeted abstention experiment	8/8 detected; precision 1.000	Unanswerable-only subset; 8 questions.
Combined reranking + abstention run	6/8 unanswerable detected	Latest combined run; do not claim abstention remained perfect in this configuration.

4. Human Validation & Cohen’s Kappa

Human labels supplied for 17 of 20 template samples. For 17 paired correctness labels, Cohen’s kappa: unweighted $\kappa = 0.184$; linear-weighted $\kappa = 0.231$; quadratic-weighted $\kappa = 0.265$. Agreement is low; results should be interpreted cautiously and the remaining template labels are incomplete. Dataset answerability metadata is not a substitute for human judgments.

5. MLflow Experiment Tracking

Experiment	Tracked runs	What this demonstrates
RBI-RAG-Evaluation (SQLite backend)	baseline_qwen; reranking_qwen; abstention_qwen; final_reranking_abstention	Four existing JSON evaluation result files imported as MLflow runs. Tracking records results; it did not rerun experiments or create GenAI traces.

6. Limitations & Reproducibility Notes

Local CPU generation is slow (prior full-RAG measurements: ~86.8 s mean without judge and ~102.5 s with judge). LLM judge uses the same qwen2.5:7b model as generation and is not an independent oracle. Refusal detection depends on lexical patterns and has configuration-dependent results. Existing report records 63/63 pytest tests passing; rerun tests if code changed after that measurement.

Demo path: show a live RBI question, the 40-question evaluation harness, baseline-versus-reranking metrics, abstention limitations, and the four MLflow runs. Git tag v1.0 is present locally; verify it is pushed to origin before submission.